
A Study of the Theoretical Foundations of Variational and Score Matching-based Diffusion Models

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1 Introduction

Score-based generative modeling and density estimation algorithms have led to the recent surge in diffusion-based generative modeling for estimating unnormalized statistical models. Unlike traditional maximum likelihood approaches, where the estimation of the (often untractable) normalizing constant restrict models' architectures and size, the core quantity of interest in score-based methods is the gradient of the log density with respect to data. Hence, the normalizing constant, which is independent of problem data, can be ignored. This allows for models with more flexible architectures and size. Score-based modeling led to state-of-the-art performance in a number of inverse problems such as compressed sensing and medical image reconstruction while novel techniques to sample from the underlying learned distribution impacted different generative tasks such as image and audio synthesis. In this project, we would like to gain a deeper understanding of the theory of score-based generative modeling. We have selected four papers which deal with two principle aspects in diffusion modeling: 1) density and score estimation: how to come up with scalable algorithms for score-based density estimation that are amenable to deep learning (i.e. large-scale network architectures and parameter space) and study their theoretical guarantees and 2) methods for efficient sampling from these models that have led to state-of-the-art results in different generative tasks. To achieve these goals, we selected few papers that delve into these two key aspects of score-based generative modeling.

2 Related Works

Density estimation of high dimensional random variables is a fundamental subject of interest in machine learning. Hence, different estimation algorithms such as Contrastive Divergence (CD) [18], Normalizing Flows [21], and Noise Contrastive Estimation (NCE) [19] etc. have been proposed. In this work, we focus on score-based density estimation algorithms which has led to recent state-of-the-art in generative modeling (i.e. stable diffusion [22]). Hyvarinen et. al [1] first proposed score-based matching as a way to model unnormalized statistical models. However, it required to estimate the trace of the hessian of the log density which can be computationally expensive. Different techniques were introduced to tackle this problem such as approximate backpropagation [23] and curvature propagation [24]. In this work, we study sliced score-based matching [2] which also aims at alleviating this problem. Another class of methods (based on variational diffusion models) aim at gradually injecting noise in training data and then learning the reverse denoisification step [3], [4] which we studied in this report. These methods have led to state-of-the-art performance in generative modelling with different modalities (images, audio [25], [26], and shapes [27]).

3 Survey of Results

3.1 Estimation of Non-Normalized Statistical Models by Score Matching

The seminal work of Hyvarinen et. al [1] introduced score-based matching as a method to estimate unnormalized statistical models. In essence, the methods can be summarized as follows: take a random vector $\mathbf{x} \in \mathbb{R}^n$. Let p_d be the “true” probability of $\mathbf{x}$ and let $p_m(\mathbf{x}, \theta) = \frac{q_m(\mathbf{x}, \theta)}{Z(\theta)}$ be the probability per our model parametrized by θ . Define the score as $s_m(\mathbf{x}; \theta) = \nabla_{\mathbf{x}} \log(p_m(\mathbf{x}, \theta))$ and then note that $s_m(\mathbf{x}; \theta) = s_d(\mathbf{x})$. Therefore, if one can define a computationally efficient algorithm for estimating θ by fitting $s_m(\mathbf{x}; \theta)$ to $s_d(\mathbf{x})$, then we gain the benefit of this estimation being independent of the intractable normalization constant. Indeed, the main observation from this paper is that the optimization cost $L(\theta)$ can be reexpressed such that we no longer require a non parametric estimate of s_d .

3.1.1 Theorem:

Assume that the model score function $s(x, \theta)$ is differentiable, as well as some weak regularity conditions. Then, the objective function:

$$L(\theta) = \frac{1}{2} \mathbb{E}_{p_d} [\|s_m(\mathbf{x}; \theta) - s_d(\mathbf{x})\|_2^2] \quad (1)$$

can be expressed as:

$$L(\theta) = \mathbb{E}_{p_d} \left[\text{tr}(\nabla_{\mathbf{x}} s_m(\mathbf{x}, \theta)) + \frac{1}{2} \|s_m(\mathbf{x}, \theta)\|_2^2 \right] \quad (2)$$

3.1.2 Proof:

Expand the square in eq. 1:

$$L(\theta) = \mathbb{E}_{p_d} \left[\frac{1}{2} \|s_d(\mathbf{x})\|_2^2 + \frac{1}{2} \|s_m(\mathbf{x}; \theta)\|_2^2 - s_d(\mathbf{x})^T s_m(\mathbf{x}; \theta) \right]$$

Ignore the term $\mathbb{E}_{p_d} [\|s_d(\mathbf{x})\|_2^2]$ because it doesn't depend on θ . It now suffices to show that: $\mathbb{E}_{p_d} [\text{tr}(\nabla_{\mathbf{x}} s_m(\mathbf{x}, \theta))] = -\mathbb{E}_{p_d} [s_d(\mathbf{x})^T s_m(\mathbf{x}; \theta)]$. Expand the inner product as a sum:

$$-\mathbb{E}_{p_d} [s_d(\mathbf{x})^T s_m(\mathbf{x}; \theta)] = -\sum_i \mathbb{E}_{p_d} [s_{i,d}(\mathbf{x})^T s_{i,m}(\mathbf{x}; \theta)]$$

Where $s_{i,d}(\mathbf{x})$ denotes the i th component of the vector $s_d(\mathbf{x})$ and similarly for $s_{i,m}(\mathbf{x}; \theta)$. Since $s_{i,d}(\mathbf{x}) = \frac{\partial \log(p_d(\mathbf{x}))}{\partial \mathbf{x}_i} = \frac{1}{p_d(\mathbf{x})} \frac{\partial p_d(\mathbf{x})}{\partial \mathbf{x}_i}$:

$$-\mathbb{E}_{p_d} [s_{i,d}(\mathbf{x})^T s_{i,m}(\mathbf{x}; \theta)] = -\int_{\mathbf{x}} p_d(\mathbf{x}) \frac{1}{p_d(\mathbf{x})} \frac{\partial p_d(\mathbf{x})}{\partial \mathbf{x}_i} s_{i,m}(\mathbf{x}; \theta) d\mathbf{x} = -\int_{\mathbf{x}} \frac{\partial p_d(\mathbf{x})}{\partial \mathbf{x}_i} s_{i,m}(\mathbf{x}; \theta) d\mathbf{x}$$

Lemma 4 in [1], proves the following partial integration trick:

3.1.3 Lemma:

Let $f(\mathbf{x})$ and $g(\mathbf{x})$ be two differentiable functions and let $\mathbf{x}_i \in \mathbb{R}$ be any index of $\mathbf{x}$, then:

$$\begin{aligned} \lim_{a \rightarrow \infty, b \rightarrow -\infty} f(\mathbf{x}_1, \dots, a, \dots, \mathbf{x}_n) g(\mathbf{x}_1, \dots, a, \dots, \mathbf{x}_n) - f(\mathbf{x}_1, \dots, b, \dots, \mathbf{x}_n) g(\mathbf{x}_1, \dots, b, \dots, \mathbf{x}_n) \\ = \int_{\mathbb{R}} f(\mathbf{x}) \frac{\partial g(\mathbf{x})}{\partial \mathbf{x}_i} d\mathbf{x}_i + \int_{\mathbb{R}} g(\mathbf{x}) \frac{\partial f(\mathbf{x})}{\partial \mathbf{x}_i} d\mathbf{x}_i \end{aligned}$$

and hence:

$$\begin{aligned} \int_{\mathbb{R}} g(\mathbf{x}) \frac{\partial f(\mathbf{x})}{\partial \mathbf{x}_i} d\mathbf{x}_i = - \int_{\mathbb{R}} f(\mathbf{x}) \frac{\partial g(\mathbf{x})}{\partial \mathbf{x}_i} d\mathbf{x}_i + \\ \lim_{a \rightarrow \infty, b \rightarrow -\infty} f(\mathbf{x}_1, \dots, a, \dots, \mathbf{x}_n) g(\mathbf{x}_1, \dots, a, \dots, \mathbf{x}_n) - f(\mathbf{x}_1, \dots, b, \dots, \mathbf{x}_n) g(\mathbf{x}_1, \dots, b, \dots, \mathbf{x}_n) \end{aligned}$$

Now let $\mathbf{x}' = \mathbf{x} \setminus \mathbf{x}_i$, $f(\mathbf{x}) = p_d(\mathbf{x})$, and $g(\mathbf{x}) = s_{i,m}(\mathbf{x}; \theta)$ then:

$$\begin{aligned} - \int_{\mathbf{x}} \frac{\partial p_d(\mathbf{x})}{\partial \mathbf{x}_i} s_{i,m}(\mathbf{x}; \theta) d\mathbf{x} &= - \int_{\mathbf{x}'} \int_{\mathbb{R}} \frac{\partial p_d(\mathbf{x})}{\partial \mathbf{x}_i} s_{i,m}(\mathbf{x}; \theta) d\mathbf{x}_i d\mathbf{x}' = \\ - \int_{\mathbf{x}'} \left[\lim_{a \rightarrow \infty, b \rightarrow -\infty} f(\mathbf{x}_1, \dots, a, \dots, \mathbf{x}_n) g(\mathbf{x}_1, \dots, a, \dots, \mathbf{x}_n) - f(\mathbf{x}_1, \dots, b, \dots, \mathbf{x}_n) g(\mathbf{x}_1, \dots, b, \dots, \mathbf{x}_n) \right] d\mathbf{x}' \\ &\quad + \int_{\mathbf{x}'} \int_{\mathbb{R}} \frac{\partial s_{i,m}(\mathbf{x}, \theta)}{\partial \mathbf{x}_i} p_d(\mathbf{x}) d\mathbf{x}' \end{aligned}$$

Since it is assumed that $p_d(\mathbf{x}_i) s_{i,m}(\mathbf{x}; \theta)$ goes to zero at infinity, we conclude that:

$$- \int_{\mathbf{x}} \frac{\partial p_d(\mathbf{x})}{\partial \mathbf{x}_i} s_{i,m}(\mathbf{x}; \theta) d\mathbf{x} = \int_{\mathbf{x}'} \int_{\mathbb{R}} \frac{\partial s_{i,m}(\mathbf{x}, \theta)}{\partial \mathbf{x}_i} p_d(\mathbf{x}) d\mathbf{x}' = \int_{\mathbf{x}} \frac{\partial s_{i,m}(\mathbf{x}, \theta)}{\partial \mathbf{x}_i} p_d(\mathbf{x}) d\mathbf{x} \quad \blacksquare$$

We implemented this method and ran experiments on the task of Multivariate Gaussian parameter estimation. We analyze the results in section 4.1

3.2 Sliced Score Matching

Although the work of Hyvärinen et al. [1] removed the need to compute or estimate the normalization constant Z_θ , the objective in eq.2 is still difficult to optimize because of the term $\nabla_{\mathbf{x}} s_m(\mathbf{x}, \theta)$ which is essentially the hessian of $\log(p_m(\mathbf{x}, \theta))$. As noted in this paper the hessian requires D times more backward passes compared to estimating the gradient and hence, is computationally expensive. This made score-based matching as presented in [1] limited to shallow models or low-dimensional data. Song et al. [2] derived an alternative objective which in essence, projects the hessian and gradient onto random vectors $\mathbf{v} \sim p_{\mathbf{v}}$ providing more suitable objectives for automatic differentiation frameworks. We next derive this objective and provide few insights into its theoretical guarantees. Drawing from the idea of the Sliced Wasserstein Distance [7], the terms in equation 2 can be projected onto random directions via multiplication with a random vector $\mathbf{v} \sim p_{\mathbf{v}}$ with $p_{\mathbf{v}}$ satisfying mild conditions (for example, $p_{\mathbf{v}}$ can be the standard normal). The new objective is then:

$$J(\theta; p_{\mathbf{v}}) = \frac{1}{2} \mathbb{E}_{\mathbf{v} \sim p_{\mathbf{v}}} \mathbb{E}_{p_d} [\|\mathbf{v}^T s_m(\mathbf{x}; \theta) - \mathbf{v}^T s_d(\mathbf{x})\|_2^2] \quad (3)$$

which can be shown to equal:

$$J(\theta; p_{\mathbf{v}}) = \mathbb{E}_{\mathbf{v} \sim p_{\mathbf{v}}} \mathbb{E}_{\mathbf{x} \sim p_d} \left[\mathbf{v}^T \nabla_{\mathbf{x}} s_m(\mathbf{x}, \theta) \mathbf{v} + \frac{1}{2} \|\mathbf{v}^T s_m(\mathbf{x}, \theta)\|_2^2 \right]$$

3.2.1 Proof

Similar to proof 3.1.2, after expanding the square, the proof boils down to showing that $\mathbb{E}_{p_{\mathbf{v}}} \mathbb{E}_{p_d} [\mathbf{v}^T \nabla_{\mathbf{x}} s_m(\mathbf{x}; \theta) \mathbf{v}] = -\mathbb{E}_{p_{\mathbf{v}}} \mathbb{E}_{p_d} [\mathbf{v}^T s_m(\mathbf{x}; \theta) \mathbf{v}^T s_d(\mathbf{x}; \theta)]$. First we can show that:

$$\begin{aligned} \mathbb{E}_{p_{\mathbf{v}}} \mathbb{E}_{p_d} [\mathbf{v}^T s_m(\mathbf{x}; \theta) \mathbf{v}^T s_d(\mathbf{x})] &= \mathbb{E}_{p_{\mathbf{v}}} \int \mathbf{v}^T s_m(\mathbf{x}; \theta) \mathbf{v}^T \nabla_{\mathbf{x}} \log(p_d(\mathbf{x})) p_d(\mathbf{x}) d\mathbf{x} \\ &= \mathbb{E}_{p_{\mathbf{v}}} \sum_{i=1}^D \int \mathbf{v}^T s_m(\mathbf{x}; \theta) v_i \frac{\partial p_d(\mathbf{x})}{\partial x_i} d\mathbf{x} \end{aligned}$$

(where we used the fact that: $\nabla_{\mathbf{x}} \log(p_d(\mathbf{x}; \theta)) p_d(\mathbf{x}) = \nabla_{\mathbf{x}} p_d(\mathbf{x})$ and then expand the inner product $\mathbf{v}^T \nabla_{\mathbf{x}} p_d(\mathbf{x})$ as a sum.) We now complete the proof by providing an alternative formulation to the one presented in [2]. We directly apply Lemma 1 with specific function substitutions. Specifically, let

$\mathbf{x}' = \mathbf{x} \setminus \{x_i\}$. Then we have:

$$\begin{aligned}
\mathbb{E}_{p_v} \sum_{i=1}^D \int \mathbf{v}^T s_m(\mathbf{x}; \theta) v_i \frac{\partial p_d(\mathbf{x})}{\partial x_i} d\mathbf{x} &= \mathbb{E}_{p_v} \sum_{i=1}^D \int \mathbf{v}^T s_m(\mathbf{x}; \theta) v_i \frac{\partial p_d(\mathbf{x})}{\partial x_i} dx_i d\mathbf{x}' \\
&= \mathbb{E}_{p_v} \sum_{i=1}^D \int \sum_{j=1}^D v_j s_{j,m}(\mathbf{x}; \theta) v_i \frac{\partial p_d(\mathbf{x})}{\partial x_i} dx_i d\mathbf{x}' \\
&= \mathbb{E}_{p_v} \sum_{i=1}^D \sum_{j=1}^D \int v_j s_{j,m}(\mathbf{x}; \theta) v_i \frac{\partial p_d(\mathbf{x})}{\partial x_i} dx_i d\mathbf{x}' \\
&= \mathbb{E}_{p_v} \sum_{i=1}^D \sum_{j=1}^D v_j v_i \int s_{j,m}(\mathbf{x}; \theta) \frac{\partial p_d(\mathbf{x})}{\partial x_i} dx_i d\mathbf{x}'
\end{aligned}$$

Now for any index pair $\{i, j\}$, we let $f(\mathbf{x}) = p_d(\mathbf{x})$ and $g(\mathbf{x}) = s_{j,m}(\mathbf{x}, \theta)$, then:

$$\begin{aligned}
&\int s_{j,m}(\mathbf{x}; \theta) \frac{\partial p_d(\mathbf{x})}{\partial x_i} dx_i + \int p_d(\mathbf{x}) \frac{\partial s_{j,m}(\mathbf{x}; \theta)}{\partial x_i} dx_i = \\
&= \lim_{a \rightarrow \infty, b \rightarrow -\infty} p_d(\mathbf{x}_1, \dots, a, \dots, \mathbf{x}_n) s_{j,m}(\mathbf{x}_1, \dots, a, \dots, \mathbf{x}_n; \theta) - p_d(\mathbf{x}_1, \dots, b, \dots, \mathbf{x}_n) s_{j,m}(\mathbf{x}_1, \dots, b, \dots, \mathbf{x}_n; \theta) \\
&- \lim_{a \rightarrow \infty, b \rightarrow -\infty} p_d(\mathbf{x}_1, \dots, a, \dots, \mathbf{x}_n) s_{j,m}(\mathbf{x}_1, \dots, a, \dots, \mathbf{x}_n; \theta) - p_d(\mathbf{x}_1, \dots, b, \dots, \mathbf{x}_n) s_{j,m}(\mathbf{x}_1, \dots, b, \dots, \mathbf{x}_n; \theta)
\end{aligned}$$

Since $p_d(\mathbf{x}) s_{j,m}(\mathbf{x}; \theta)$ goes to zero at infinity:

$$\int s_{j,m}(\mathbf{x}; \theta) \frac{\partial p_d(\mathbf{x})}{\partial x_i} dx_i + \int p_d(\mathbf{x}) \frac{\partial s_{j,m}(\mathbf{x}; \theta)}{\partial x_i} dx_i = 0$$

Hence:

$$\begin{aligned}
\mathbb{E}_{p_v} \sum_{i=1}^D \int \mathbf{v}^T s_m(\mathbf{x}; \theta) v_i \frac{\partial p_d(\mathbf{x})}{\partial x_i} d\mathbf{x} &= -\mathbb{E}_{p_v} \sum_{i=1}^D \sum_{j=1}^D v_i v_j \int p_d(\mathbf{x}) \frac{\partial s_{j,m}(\mathbf{x}; \theta)}{\partial x_i} dx_i d\mathbf{x}' = \\
&= -\mathbb{E}_{p_v} \int p_d(\mathbf{x}) \sum_{i=1}^D \sum_{j=1}^D v_i v_j \frac{\partial s_{j,m}(\mathbf{x}; \theta)}{\partial x_i} dx_i d\mathbf{x}' \\
&= -\mathbb{E}_{\mathbf{v} \sim p_v} \mathbb{E}_{\mathbf{x} \sim p_d} [\mathbf{v}^T \nabla_{\mathbf{x}} s_m(\mathbf{x}, \theta) \mathbf{v}] \quad \blacksquare
\end{aligned}$$

3.2.2 Variance Reduced Empirical Loss

The empirical loss term for sliced score-based matching is:

$$\hat{J}(\theta, \mathbf{x}_1^N, \mathbf{v}_{11}^N) = \frac{1}{N} \frac{1}{M} \sum_{i=1}^M \sum_{j=1}^N \mathbf{v}_{ij}^T \nabla_{\mathbf{x}} s_m(\mathbf{x}_i, \theta) \mathbf{v}_{ij} + \frac{1}{2} (\mathbf{v}_{ij}^T s_m(\mathbf{x}_i; \theta))^2 \quad (4)$$

where $\mathbf{x}_1^N$ refers to a dataset with N datapoints. $\mathbf{v}_{11}^N$ is the collection of sampled random vectors where M vectors are sampled for each datapoint (In practice $M = 1$, i.e. we perform a single random projection). If the p_v is the multivariate standard normal or multivariate Rademacher distribution then: $\mathbb{E}_{p_v}[(\mathbf{v}^T s_m(\mathbf{x}, \theta))^2] = \|s_m(\mathbf{x}, \theta)\|_2^2$. Hence, the second term in eq. 4 can be integrated analytically resulting in a variance reduced loss term:

$$\hat{J}_{\text{vr}}(\theta, \mathbf{x}_1^N, \mathbf{v}_{11}^N) = \frac{1}{N} \frac{1}{M} \sum_{i=1}^M \sum_{j=1}^N \mathbf{v}_{ij}^T \nabla_{\mathbf{x}} s_m(\mathbf{x}_i, \theta) \mathbf{v}_{ij} + \frac{1}{2} \|s_m(\mathbf{x}, \theta)\|_2^2 \quad (5)$$

3.2.3 Theoretical Analysis

We now summarize a few theoretical properties of the loss in eq. 4.

I) Consistency: Let θ^* be the true parameter of the data distribution. Then for ever $M > 0$: $\hat{\theta}_{N,M} \rightarrow \theta^*$, $N \rightarrow \infty$

Proof sketch First prove that for the loss in eq. 3: $J(\theta; p_{\mathbf{v}}) = 0 \Leftrightarrow \theta = \theta^*$ by expanding the inner terms and noting that $s_m(\mathbf{x}; \theta) = s_d(\mathbf{x}) \Leftrightarrow \log p_m(\mathbf{x}, \theta) = \log p_d(\mathbf{x}) + C$. Then conclude using the identifiability assumption. Second prove the uniform convergence of $\hat{J}(\theta, \mathbf{x}_1^N, \mathbf{v}_{11}^N)$. In other words that:

$$\mathbb{E}_{p_{\mathbf{v}}, p_d} \left[\sup_{\theta \in \Theta} |\hat{J}(\theta, \mathbf{x}_1^N, \mathbf{v}_{11}^N) - J(\theta, p_{\mathbf{v}})| \right] \leq O \left(\text{diam}(\Theta) \sqrt{\frac{D}{N}} \right)$$

Where D is the dimension of the parameter space Θ . Combining these two facts, the consistency of the estimator follows.

II) Asymptotic Normality If $p_{\mathbf{v}}$ is the multivariate Rademacher distribution, then: $\sqrt{N}(\hat{\theta}_{N,M} - \theta^*)$ converges in distribution to $\mathcal{N}(0, \Sigma)$ where Σ is defined in Appendix B.4 in [2]. The proof is similar to the proof of asymptotic convergence of the MLE (See [8]).

We reproduced some results from [2] on the CelebA dataset and provide an analysis in section 4.2.

3.3 Variational Diffusion Models

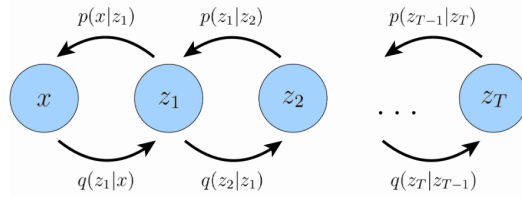


Figure 1: Figure borrowed from [9]. An MHVAE. The generative process is such that each latent z_{i+1} is generated only from the previous latent z_i (Markov assumption).

We study another class of methods to train score-based networks which fall under the umbrella of *Variational Diffusion Models (VDM)*. These methods have interesting resemblance to Markovian Hierarchical Variational Autoencoder (MHVAE) [14], [15] under some restrictions. We first provide a brief summary of MHVAEs (for a more detailed review of VAEs and MHVAEs, we refer the reader to [9]). In MHVAEs, we model the joint probability of data and latent variables as:

$$p(\mathbf{x}, \mathbf{z}_{1:T}) = p(\mathbf{z}_T) p_{\theta}(\mathbf{x} | \mathbf{z}_1) \prod_{t=2}^T p_{\theta}(\mathbf{z}_{t-1} | \mathbf{z}_t)$$

where this factorization is true under the Markov assumption. The posterior is expressed as:

$$q_{\phi}(\mathbf{z}_{1:T} | \mathbf{x}) = q_{\phi}(\mathbf{z}_1 | \mathbf{x}) \prod_{t=2}^T q_{\phi}(\mathbf{z}_t | \mathbf{z}_{t-1})$$

where p_{θ} and q_{ϕ} are the variational approximation (learned functions parametrized by ϕ and θ) of the true distributions $p(\mathbf{z}_{t-1} | \mathbf{z}_t)$ and $q(\mathbf{z}_t | \mathbf{z}_{t-1})$. Then the distribution of observed data $p(\mathbf{x})$ can be modelled by maximizing the following ELBO:

$$E_{q_{\phi}(\mathbf{z}_{1:T} | \mathbf{x})} \left[\frac{p(\mathbf{z}_T) p_{\theta}(\mathbf{x} | \mathbf{z}_1) \prod_{t=2}^T p_{\theta}(\mathbf{z}_{t-1} | \mathbf{z}_t)}{q_{\phi}(\mathbf{z}_1 | \mathbf{x}) \prod_{t=2}^T q_{\phi}(\mathbf{z}_t | \mathbf{z}_{t-1})} \right] \quad \mathbf{z}_T \sim \mathcal{N}(0, I) \quad (6)$$

With VDMs, we restrict the latent variables dimension to equal that of the data. Hence, the latent distributions x_t are defined over the same space as the data x_0 . In addition the latent encoders q_{ϕ} are assumed to belong to a much more restrictive class of functions (linear Gaussians with possibly learned parameters. Hence the forward pass can be regarded as gradual noisification and the backward steps as denoising steps). VDMs can be trained by maximizing the ELBO in 6. The restrictions above however, allows us to rewrite the ELBO itself in terms of several components:

$$\begin{aligned} \log(p(\mathbf{x})) &\geq \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log(p_\theta(\mathbf{x}_0|\mathbf{x}_1))] - \mathbb{E}_{q(\mathbf{x}_{T-1}|\mathbf{x}_0)} \text{KL}(q(\mathbf{x}_T|\mathbf{x}_{T-1}) \parallel p(\mathbf{x}_T)) \\ &\quad - \sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_{t+1}, \mathbf{x}_{t-1}|\mathbf{x}_0)} [\text{KL}(q(\mathbf{x}_t|\mathbf{x}_{t-1}) \parallel p_\theta(\mathbf{x}_t|\mathbf{x}_{t-1}))] \end{aligned}$$

which, by noting that $q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0) = \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)q(\mathbf{x}_t, \mathbf{x}_0)}{q(\mathbf{x}_{t-1}, \mathbf{x}_0)}$, can be rewritten in a form where all expectations depend only on one variable.

$$\begin{aligned} \log(p(\mathbf{x})) &\geq \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log(p_\theta(\mathbf{x}_0|\mathbf{x}_1))] - \text{KL}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T)) \\ &\quad - \sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [\text{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))] \end{aligned} \quad (7)$$

The process of training the ELBO in eq.7 is illustrated in fig. 2

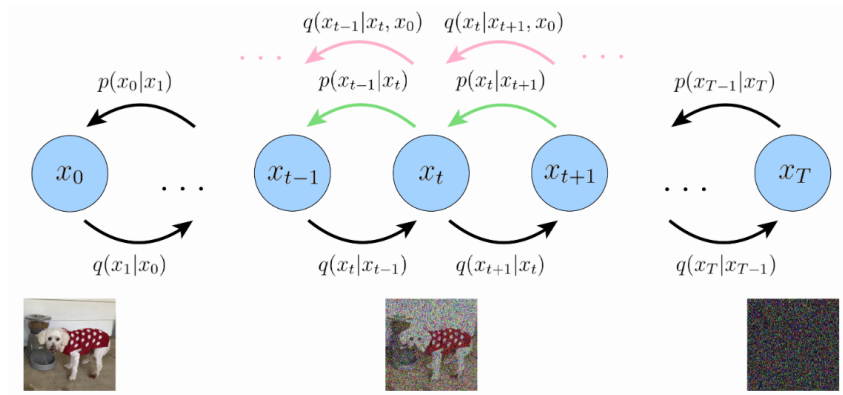


Figure 2: Figure borrowed from [9]. The various components involved with training a Variational Diffusion Model corresponding to the ELBO in eq. 7

Now, if we examine the RHS, we see that in order to make this optimization tractable, we need an efficient way to sample from $q(\mathbf{x}_{t+1}|\mathbf{x}_t)$, $q(\mathbf{x}_t|\mathbf{x}_0)$ as well as way to compute the KL term under the sum. We can here use our Gaussian assumption to derive the following tractable expressions:

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \sqrt{\alpha_t}\mathbf{x}_{t-1}, (1 - \alpha_t)\mathbf{I}) \quad (8)$$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_q(\mathbf{x}_t, \mathbf{x}_0), \Sigma_q(t)) \quad (9)$$

$$\begin{aligned} \text{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)) &= \text{KL}(\mathcal{N}(\mathbf{x}_{t-1}; \mu_q, \Sigma_q(t)) \parallel \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta, \Sigma_q(t))) \\ &= \frac{1}{2\sigma_q^2(t)} \frac{\bar{\alpha}_{t-1}(1 - \alpha_t)^2}{(1 - \bar{\alpha}_t)^2} \|\hat{\mathbf{x}}_\theta(\mathbf{x}_t, t) - \mathbf{x}_0\|_2^2 \end{aligned} \quad (10)$$

where α_t, α_{t-1} are coefficients controlling the parameters of a Gaussian, μ_q and Σ_q are tractable expressions for the mean and covariance normal distribution (see [9] for their detailed expressions) and finally, $\hat{\mathbf{x}}_\theta(\mathbf{x}_t, t)$ is a learnable function which tries to predict the data sample $\mathbf{x}_0$ from $\mathbf{x}_t$ at time t . (Note that in eq. 10, we use the closed-form solution for the KL divergence between two Gaussians). Hence, the ELBO in 7 can be maximized via regular Monte Carlo optimization.

So far, we focused on studying methods for *training* score-based networks. (Note that the VDM approach above can be easily extended to learning score functions). We now summarize few methods for *generating* samples from the learned score functions. Interestingly, the view of VDMs as some restricted class of MHVAEs elucidates the sampling procedures which underlies many state-of-the-art methods for score-based diffusion model generative modelling.

3.4 Score-Based Diffusion Models

3.4.1 Generative Modeling by Estimating Gradients of the Data Distribution [3]

In the work of Song et al. [3], a generative modeling technique is presented which produces samples from the data distribution via Langevin dynamics. This procedure only requires the score function which can be estimated using techniques such as sliced score matching. However, the authors explain why a naive application of this idea may be challenging (manifold hypothesis, low density regions etc.) and propose to learn and sample from *Noise Conditional Score Networks* instead.

Noise Conditional Score Networks (NCSN) Define a sequence of noise levels $\{\sigma_i\}_{i=1}^L$ such as $\sigma_i > \sigma_{i+1} \forall i \in [1, L-1]$ and perturb each data sample by adding an independent normal noise value with variance σ_i to obtain the perturbed data distribution:

$$q_{\sigma_i}(\mathbf{x}) = \int p_{\text{data}}(\mathbf{t}) \mathcal{N}(\mathbf{x}|\mathbf{t}, \sigma^2 I)$$

Then an NCSN can be trained using sliced score based matching to jointly estimate the scores of all perturbed distributions mitigating the problems described above:

$$s_{\theta}(\mathbf{x}, \sigma) \approx \nabla_{\mathbf{x}} \log(q_{\sigma_i}(\mathbf{x}))$$

NCSN inference via annealed Langevin dynamics After training the NCSN, samples can be generated using annealed Langevin dynamics (Algorithm. 1) which follows a procedure inspired by simulated annealing. In short, we run langevin dynamics to sample a point from distribution $q_{\sigma_i}(\mathbf{x})$ and then starting from this point, run Langevin dynamics again to sample from $q_{\sigma_{i+1}}(\mathbf{x})$. We do this from $\forall i$. The final sample from $q_{\sigma_T}(\mathbf{x})$ approximates a sample from the target distribution p_{data} .

Algorithm 1 Annealed Langevin dynamics.

Require: $\{\sigma_i\}_{i=1}^L, \epsilon, T$.

- 1: Initialize $\tilde{\mathbf{x}}_0$
- 2: **for** $i \leftarrow 1$ to L **do**
- 3: $\alpha_i \leftarrow \epsilon \cdot \sigma_i^2 / \sigma_L^2$ $\triangleright \alpha_i$ is the step size.
- 4: **for** $t \leftarrow 1$ to T **do**
- 5: Draw $\mathbf{z}_t \sim \mathcal{N}(0, I)$
- 6: $\tilde{\mathbf{x}}_t \leftarrow \tilde{\mathbf{x}}_{t-1} + \frac{\alpha_i}{2} \mathbf{s}_{\theta}(\tilde{\mathbf{x}}_{t-1}, \sigma_i) + \sqrt{\alpha_i} \mathbf{z}_t$
- 7: **end for**
- 8: $\tilde{\mathbf{x}}_0 \leftarrow \tilde{\mathbf{x}}_T$
- 9: **end for**
- return** $\tilde{\mathbf{x}}_T$

Figure 3: Annealed Langevin Dynamics. Figure borrowed from [3]

3.4.2 Score-Based Generative Modeling through Stochastic Differential Equations [4]

Song et al. in [3] proposed to perturb the data with a specific number of noise scales (i.e. annealing) to deal with certain problems related to training score-based generative models (i.e low density regions in the data distribution). In [4], the authors consider a continuous time diffusion process consisting of two parts. 1) continuously infusing data with noise to smoothly transition from the data to a noise distribution (Forward SDE) and then train the network to generate samples from the data distribution by first sampling from the noise distribution and running a reverse SDE operation. Specifically, the forward diffusion process is modeled as the solution to the following SDE:

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

where $\mathbf{w}$ is the standard Wiener process, $\mathbf{f}(\cdot, t) : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is the drift coefficient of $\mathbf{x}(t)$ and $g(\cdot) : \mathbb{R} \rightarrow \mathbb{R}$ is the diffusion coefficient of $\mathbf{x}(t)$. The reverse process is shown to also be a diffusion process in time which is given by the following reverse-time SDE:

$$d\mathbf{x} = [\mathbf{f}(\mathbf{x}, t) - g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x})]dt + g(t)d\tilde{\mathbf{w}}$$

where dt is an infinitesimal negative timestep. The scores $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ can be estimated by training score-based model using the methods in [1], [2], or [17].

4 Experiments

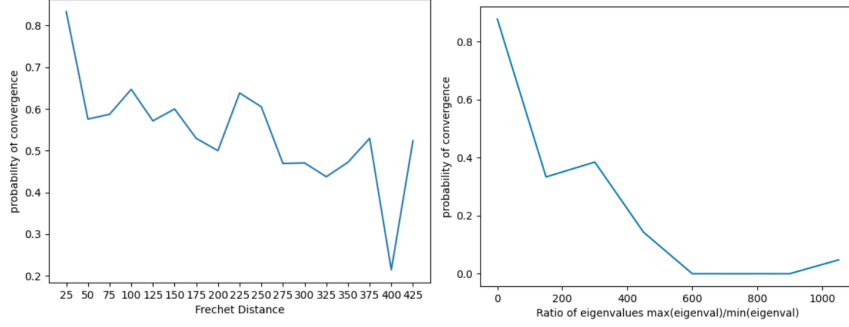


Figure 4: (a) Probability of convergence as a function of Frechet Distance between target and initial guess of the distribution (b) Probability of convergence as a function of the ratio of the max and small eigenvalues

The code for the following results can be found here: <https://github.com/mqadri9/Score-based-Matching-Density-Estimation>

4.1 Multivariate Gaussian Parameter Estimation

We begin by reproducing the results from Hyvarinen et al. [1] aimed at estimating the parameters of a multivariate Gaussian. The probability density function is given by:

$$p(\mathbf{x}, \Sigma, \mu) = \frac{q(\mathbf{x}, \Sigma, \mu)}{Z(\mu, \Sigma)} = \frac{\exp(-\frac{1}{2}(\mathbf{x} - \mu)^T \Sigma^{-1}(\mathbf{x} - \mu))}{(2\pi)^{\frac{k}{2}} \det(\Sigma)^{\frac{1}{2}}} \quad (11)$$

We sample N points from a target d -dimensional Gaussian distribution $p(\mathbf{x}, \Sigma, \mu)$. Our objective is to recover the true parameters of p from these samples. Our optimization variables are the mean μ (d parameters) and $\tilde{\Phi}$ ($\frac{d^2-d}{2}$ parameters) where $\tilde{\Phi}^T \tilde{\Phi} = \tilde{\Sigma}$ with $\tilde{\Sigma}$ being the estimated covariance. This parametrization of the covariance constraints it to remain positive definite with high probability throughout the optimization procedure. We use the score-based matching loss function 2 where the gradient and hessian are computed analytically:

$$\nabla_{\mathbf{x}} \log(\tilde{q}(\mathbf{x}, \Sigma, \mu)) = -(\mathbf{x} - \mu)^T \Sigma^{-1} \quad (12)$$

$$\nabla_{\mathbf{x}}^2 \log(\tilde{q}(\mathbf{x}, \Sigma, \mu)) = -\Sigma^{-1} \quad (13)$$

$\tilde{q}$ being the estimated unnormalized probability density.

4.1.1 Empirical Analysis

We noted that the algorithm (per our implementation) may empirically fail to converge (after $10e^4$ iteration) for specific target distributions p and initialization of the parameters $\tilde{\mu}$ and $\tilde{\Phi}$. We empirically analyze this behavior based on the following two hypothesis:

- The probability of convergence depends on the distance between the target distribution and our initial guess. We used the Frechet distance as a measure of distance between two Gaussians:

$$d^2 = |\mu_p - \mu_{\tilde{p}}|^2 + \text{tr}(\Sigma_p - \Sigma_{\tilde{p}} - 2(\Sigma_p \Sigma_{\tilde{p}})^{\frac{1}{2}})$$

- The probability of convergence depends on the variability of the sampled data across its principle axes which can be estimated by the the ratio $\frac{\lambda_{\max}}{\lambda_{\min}}$. $\lambda_{\max}$ and $\lambda_{\min}$ are respectively the maximum and minimum eigenvalues obtained from the spectral decomposition of the target covariance Σ . Note that this ratio also equals the condition number $\text{Cond}(\Sigma)$.

We perform ~ 500 runs of the algorithm with randomized target and initial parameters for the 2D Gaussian case (we verified that same observations can be made for the higher dimensional 5D

Gaussian case) and plot the probability of convergence after N iterations as a function of the Frechet distance (figure (b)) and the ratio of eigenvalues (figure (c)). The probability of convergence P_c is defined as:

$$P_c = \frac{\text{\#Runs}[\epsilon_1 < x < \epsilon_2 \ \& \ \text{Converged}=\text{True}]}{\text{\#Runs}[\epsilon_1 < x < \epsilon_2]} \quad (14)$$

where ϵ_1, ϵ_2 are the lower and upper bound of either the Frechet distance or ratio of eigenvalues in a specific interval.

We note that there is no correlation between the Frechet distance and P_c . However, P_c is highly correlated with the spread of the samples along its principal components: as $\text{Cond}(\Sigma) = \frac{\lambda_{\max}}{\lambda_{\min}}$ increases, the true covariance matrix becomes increasingly ill-conditioned which is known to be difficult to estimate using MLE [5], and as we see here, also for score-based matching.

4.2 Experiments with Sliced Score Matching (SSM)

We reproduced the results from [2] on the CelebA and report the Frechet Inception Distance (FID) in table I for various architectures/losses at different training steps.

VAE We trained variational autoencoder (VAEs) using Sliced Score Based Matching (SSM) and compared it against training via maximization of the Variational Evidence Lower Bound (ELBO) as well as optimizing using the Stein and Spectral estimators.

WAE We trained Wasserstein Auto-Encoders (WAEs) using Sliced Score Based Matching (SSM) and compared the results against training using the Stein and Spectral estimators.

Table I: FID scores of different methods versus number of training iterations on CelebA dataset.

	Method \ Iteration	10k	40k	70k	100k
VAE	ELBO	89.378	76.553	74.918	73.165
VAE	Stein	103.491	92.001	96.717	99.137
VAE	Spectral	99.82	93.690	99.856	101.582
VAE	SSM	84.259	71.000	69.034	67.605
WAE	Stein	81.544	69.795	66.549	64.227
WAE	Spectral	82.501	69.863	66.722	65.587
WAE	SSM	83.039	68.293	67.709	63.559

4.2.1 Discussion

The results in table I are quite comparable to the official results reported in table 3 in [2]. We note that training using SSM outperforms other methods for both the VAE and WAE architectures after 100k training steps. However, the results with SSM are still far worse than the current state-of-the-art results in generative modeling on CelebA which are generally trained using diffusion model algorithms. Hence, the process of learning hierarchical representations through gradual noisification and denoisification procedures is playing an important role in making score-based diffusion models the state-of-the-art algorithms in generative modeling.

5 Conclusion

In this project, we explored several aspects pertaining to score-based generative modelling. We first, studied loss functions that made score-based matching tractable and amenable to optimization using modern autodiff engines. We then looked into connections between diffusion models and variational inference and finally, examined state-of-the-art mechanisms to sample from these networks.

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